TVRTKO ŠAPINA

Third-year Mathematics student interested in quantitative finance, mathematical modelling and machine learning, with hands-on experience in Python for data science, automation, and financial analysis.

EDUCATION

University of Zagreb, Faculty of Science | BSc in Mathematics · *Expected 2027*

Relevant coursework: Probability, Statistics, Numerical Mathematics, Linear Algebra, Calculus, Differential Equations, Measure Theory, Data Structures and Algorithms

PROJECTS (Code available on GitHub)

Python Backtesting & Risk Engine | Python, pandas, NumPy, SciPy, statsmodels

- Built an event-driven backtesting engine that turns OHLCV data into portfolios, aligns trading calendars across exchanges, executes at the next day's open to avoid look-ahead bias, and models transaction costs and short selling.
- Implemented CAGR, Sharpe, Sortino, Calmar and max drawdown; historical, parametric and Monte Carlo (normal / Student-t) VaR and Expected Shortfall; risk attribution and hypothetical/historical stress testing.
- Tested mean-reversion and cointegration-based pairs trading on eight US and European large caps, using in-sample parameter selection and out-of-sample evaluation against equal-weight and minimum-variance benchmarks.

Option Pricing Toolkit | Python, numerical methods

- Developed a toolkit for European, American and Bermudan call and put options using CRR binomial trees, Black–Scholes and Monte Carlo; added Greeks for all three pricing methods.
- Implemented Euler–Maruyama, tamed Euler and Milstein schemes for stochastic differential equations. Live demo: optionpricingtoolkit-dewbzm3qzn6d2rh3caof8n.streamlit.app

Banknote Authentication | Python, scikit-learn

- Preprocessed data, explored clustering, and compared logistic regression, SVM, decision trees and gradient boosting classifiers.

EXPERIENCE

IC Academy | InterCapital · *Sep 2026*

- Selected through an asset-management assignment organized by a leading independent investment company in Southeast Europe. The program offered a comprehensive introduction to InterCapital's business, including macroeconomic analysis, valuation of bonds, equities and real estate, as well as trading.

Student Data Scientist | Hrvatski Telekom · *Jul-Sep 2026*

- Performed feature engineering for a customer credit-risk assessment model and extracted relevant data from SQL databases.

Controlling & Risk Management Intern | E.ON · *Mar-Jul 2025*

- Built Python tools to automate HUDEX market-data extraction and estimate monthly natural-gas prices from quarterly and annual forwards using polynomial interpolation.

Photography and digital marketing - freelance projects | Nu bar, Nec, Villa Madomi · *2022-2025*

ACTIVITIES & COURSES

Financijski klub — Member, Algotrading and Market Modelling Group | Akuna Capital Options 101 course

TECHNICAL SKILLS & LANGUAGES

Python (pandas, NumPy, SciPy, statsmodels, scikit-learn), C, basic C++, SQL, Excel | Croatian (native), English (C1), German (A1)